



Published in final edited form as:

J Magn Reson Imaging. 2012 November ; 36(5): 1011–1014. doi:10.1002/jmri.23741.

Proton Density Fat-Fraction: A Standardized MR-Based Biomarker of Tissue Fat Concentration

Scott B. Reeder, MD, PhD,

Departments of Radiology, Medical Physics, Biomedical Engineering, and Medicine, University of Wisconsin, Madison, Wisconsin, USA

Houchun H. Hu, PhD, and

Department of Radiology, Children's Hospital Los Angeles, University of Southern California, Los Angeles, California, USA

Claude B. Sirlin, MD

Liver Imaging Group, Department of Radiology, University of California, San Diego, California, USA

On February 19–22, the 2012 ISMRM Workshop on Fat–Water Separation was held in Long Beach, California. Over 130 attendees from 13 countries convened to discuss historical aspects of water–fat imaging, as well as technical and clinical developments, and remaining challenges. Among these attendees were new and established investigators. A comprehensive overview of the meeting and presentations from the 25 invited speakers at the workshop was recently published in *Magnetic Resonance in Medicine* (1).

As illustrated in Figure 1 below, scientific interest in fat quantification using MR spectroscopy (MRS) and MR imaging (MRI) has grown steadily in the past decade. Accordingly, many of the invited lectures and scientific abstracts at the meeting focused on emerging MRS- and MRI-based methods to quantify tissue fat concentration. Applications included quantification of tissue fat concentration in organs and tissues such as liver, pancreas, skeletal muscle, heart, thymus, bone marrow, and white and brown adipose tissue, as well as the use of MRI techniques to quantify and characterize fat content for the food industry. During these lectures and ensuing discussions, a theme that emerged multiple times was the need to standardize MR-based biomarkers for assessing tissue fat concentration.

While many different MR-based biomarkers may be considered for fat quantification, an informal consensus was reached at the meeting that proton-density fat-fraction (PDFF) is currently the most practical and meaningful MR-based biomarker for this purpose. Thus, the goal of this editorial is to provide a formal statement from three of the organizers of the ISMRM workshop on the choice of PDFF as the standardized MR-based biomarker for tissue fat concentration.

It is our position that PDFF is the most appropriate standardized biomarker of tissue fat concentration when measured appropriately with quantitative MRS or MRI techniques. In the remainder of this editorial we will define PDFF, justify its use as a standardized MR-based biomarker of tissue fat concentration, explain current technical approaches to measure

PDFF correctly, and explain differences of PDFF with tissue-based measures of fat concentration such as biochemical assays of tissue fat.

In general, several important metrics of a valid biomarker must be considered when choosing a biomarker for standardization. In addition to measuring a clinically relevant quantity (e.g., liver fat content), the biomarker must be accurate, precise, robust, and reproducible. *Accuracy* reflects the degree of correlation between the biomarker and an accepted reference standard, i.e., how well does the biomarker predict the quantity being measured? *Precision*, or repeatability, reflects the variability of the biomarker when repeated measurements are made, including within an examination and between examinations. Precision is particularly important for treatment monitoring and for longitudinal measurements. An imaging biomarker is considered *robust* if it is insensitive to changes in the scanning parameters that might be encountered in clinical practice or in a clinical trial. Finally, validation studies should be *reproducible*, with low variability across platform, scanner manufacturer, field strength, reader, reading center, and technologist. As we discuss below, the PDFF is an imaging biomarker that best meets all of these criteria.

Standardization of all biomarkers, including PDFF, is critical for research and clinical purposes. This permits pooling of results across multiple centers, including meta-analyses and multi-center studies, as well as implementation into widespread clinical practice and acceptance. Finally, standardization of PDFF as the MR-based biomarker of tissue fat concentration would permit portability of tissue fat concentration measurements for patients being cared for at multiple institutions.

Prototype implementations of MR-based measurements of PDFF proposed by multiple groups have also demonstrated a practical strength of PDFF: the ability to record PDFF values directly from PDFF maps (2–4). Fully automated calculation of PDFF maps generated using imaging-based methods or PDFF values measured using automated postprocessing of single-voxel MRS-based methods are highly practical and easily accepted by clinicians (5). While the reconstruction algorithms are advanced and require complex computations, the algorithms can be implemented seamlessly to rapidly generate MRI-based PDFF maps or MRS-based PDFF values online. So long as the underlying complexity is invisible to the practitioner, the use of PDFF algorithms will be easily adopted and accepted in a clinical setting. This practical aspect of PDFF is understated, and is a clear advantage over other methods, including calculation of fat-fraction from conventional in/opposed phase imaging, which requires the user to record signal intensity values recorded in co-localized regions on images acquired at different echo times and then calculate an approximate fat-fraction (6).

Several quantitative MRS-based (5,7–15) and MRI-based (2–4,7,8,11,14,16–36) methods have recently been developed for quantification of tissue triglyceride content. Both MRS- and MRI-based methods share the same underlying MR physics concepts, exploiting differences in resonance frequencies (“chemical shift”) between water and fat proton signals. These methods first decompose the signals attributable to water and fat, and then normalize these signals by calculating the ratio of fat signal divided by the total signal from water and fat, i.e., signal fat-fraction. The fat–water signal ratio, which is the ratio of fat

signal to the water signal, is an alternative metric. An important property of voxel-wise signal fat-fraction (and fat-water signal ratio) measurements, whether MRI- or MRS-based, is that they are independent of RF coil sensitivity (“B₁” sensitivity) or any other signal-intensity scaling factors that affect the voxel. This property eliminates errors introduced by inhomogeneous coil sensitivity that may occur when muscle or spleen is used as an internal normalization measurement.

Unfortunately, many biological, physical, and technical factors affect the relative MR signals from fat and water and so may confound the accuracy of signal fat-fraction as a reliable metric of fat content within tissue. To date, several specific confounding factors have been identified and solutions to mitigate or correct for these factors have been developed. Using magnitude-based quantitative MRI methods (i.e., those that discard phase from the signal), the effects of T₁ related bias (17,25), T₂^{*} decay (17,35,36), and spectral complexity of fat (17,36), must be considered to quantify fat. Complex-based MRI methods (i.e., those methods that preserve both signal phase and magnitude) must also consider the effects of noise related bias (25) and of eddy currents (37,38), as well as the effects of gradient timing misregistrations and other phase errors that manifest when using bipolar readout acquisitions (39). MRS methods, like MRI methods, must also address the effects of T₁ related bias and spectral complexity of fat; they additionally must consider the effects of T₂ decay and J-coupling, as well as the dependency of these latter effects on the type of MRS method used (e.g., STEAM versus PRESS) (9).

If all confounding factors have been addressed or mitigated, the signal fat-fraction is equivalent to the proton density fat-fraction (PDFF), which is **defined** as the ratio of density of mobile protons from fat (triglycerides) and the total density of protons from mobile triglycerides and mobile water. The PDFF is a fundamental property of tissue and reflects the concentration of fat within that tissue. While it is conceivable that additional confounding factors may be identified, such confounding factors are likely to be minor.

To date, considerable work has focused on optimization of acquisition parameters to maximize noise performance, accuracy, and precision of PDFF measurements over the entire biological range of fat content (36–38,40). Ongoing technical refinements are likely to further reduce the impact of confounding factors—those currently known as well as possible additional ones yet to be identified—and further improve PDFF measurement performance.

It should be noted that while PDFF closely correlates with tissue triglyceride concentration measured using standard biochemical assays that measure fat mass-fraction, PDFF is not equivalent to the mass fat-fraction. Tissue contains a significant quantity of NMR invisible material that is not measured using MR-based methods leading to natural disagreement between PDFF and mass fat-fraction (41,42). Fortunately, the proton densities of water and triglyceride are almost identical (42). For this reason, in phantom studies where the phantom is compromised entirely of water and fat, the PDFF and mass fat-fraction are equivalent. Excellent agreement between MR-based PDFF and ground truth mass fat-fraction correlate closely and also demonstrate excellent agreement (41,43). In tissue, however, where a significant proportion of tissue is NMR invisible, agreement between mass fat-fraction and PDFF is neither expected nor observed (43–45). Conversion between mass fat-fraction and

PDFF is possible if the NMR invisible content of tissue is known (41). Using currently techniques, however, the NMR invisible content of tissue cannot be measured reliably in vivo, and conversion from PDFF to mass fat-fraction requires assumptions that may introduce measurement error. For these reasons, we argue that, at present, conversion from PDFF to mass fat-fraction currently is neither practical nor necessary.

PDFF is an unconfounded and fundamental property of tissue, and it is a practical and useful measurement of tissue fat concentration. For this reason, we propose that PDFF become the standardized MR-based biomarker of tissue fat concentration when measured by appropriate quantitative MRS or MRI methods. In addition to being accurate (2–4,46) and precise (46), PDFF is a *robust* quantitative biomarker because it is insensitive to changes in acquisition parameters. This leads the PDFF to become a *reproducible* quantitative biomarker, stable across scanner platform, scanner manufacturer, imaging center, and even field strength. A standardized biomarker must be both robust and reproducible, and PDFF is the best-suited quantitative MR-based biomarker of tissue fat concentration to meet these requirements. Without correction for confounding factors, the signal fat-fraction is neither robust nor reproducible, and the uncorrected signal fat-fraction should not be used routinely for quantification of tissue fat concentration.

In summary, it is our position that PDFF is the best current metric for a standardized MR-based biomarker of tissue fat concentration. It represents a fundamental property of tissue that, if measured properly, accurately reflects the concentration of fat in tissue. Because it quantifies a fundamental property of tissue, it is independent of acquisition parameters, platform, scanner manufacturer, imaging center and field strength. These features lend naturally to standardization, which is critical for large-scale research endeavors and widespread clinical implementation. PDFF is also a highly practical biomarker that is straightforward and simple to implement clinically, features essential for rapid and widespread clinical adoption. Given the rapid emergence of MR-based methods for quantifying tissue fat concentration (Fig. 1), we believe that use of a standardized MR-based biomarker of tissue fat concentration is needed urgently, and that PDFF is the best biomarker to meet this need.

REFERENCES

1. Hu H, Börnert P, Hernando D, et al. ISMRM Workshop on fat-water separation: insights, applications & progress in MRI. *Magn Reson Med*. 2012 (in press).
2. Meisamy S, Hines CD, Hamilton G, et al. Quantification of hepatic steatosis with T1-independent, T2-corrected MR imaging with spectral modeling of fat: blinded comparison with MR spectroscopy. *Radiology*. 2011; 258:767–775. [PubMed: 21248233]
3. Yokoo T, Bydder M, Hamilton G, et al. Nonalcoholic fatty liver disease: diagnostic and fat-grading accuracy of low-flip-angle multiecho gradient-recalled-echo MR imaging at 1.5 T. *Radiology*. 2009; 251:67–76. [PubMed: 19221054]
4. Yokoo T, Shieh-morteza M, Hamilton G, et al. Estimation of hepatic proton-density fat fraction by using MR imaging at 3.0 T. *Radiology*. 2011; 258:749–759. [PubMed: 21212366]
5. Pineda N, Sharma P, Xu Q, Hu X, Vos M, Martin DR. Measurement of hepatic lipid: high-speed T2-corrected multiecho acquisition at 1H MR spectroscopy—a rapid and accurate technique. *Radiology*. 2009; 252:568–576. [PubMed: 19546430]

6. Reeder SB, Cruite I, Hamilton G, Sirlin CB. Quantitative assessment of liver fat with magnetic resonance imaging and spectroscopy. *J Magn Reson Imaging*. 2011; 34 spcone.
7. Cowin GJ, Jonsson JR, Bauer JD, et al. Magnetic resonance imaging and spectroscopy for monitoring liver steatosis. *J Magn Reson Imaging*. 2008; 28:937–945. [PubMed: 18821619]
8. d'Assignies G, Ruel M, Khiat A, et al. Noninvasive quantitation of human liver steatosis using magnetic resonance and bioassay methods. *Eur Radiol*. 2009; 19:2033–2040. [PubMed: 19280194]
9. Hamilton G, Middleton MS, Bydder M, et al. Effect of PRESS and STEAM sequences on magnetic resonance spectroscopic liver fat quantification. *J Magn Reson Imaging*. 2009; 30:145–152. [PubMed: 19557733]
10. Hamilton G, Yokoo T, Bydder M, et al. In vivo characterization of the liver fat (1)H MR spectrum. *NMR Biomed*. 2011; 24:784–790. [PubMed: 21834002]
11. Kim H, Taksali SE, Dufour S, et al. Comparative MR study of hepatic fat quantification using single-voxel proton spectroscopy, two-point dixon and three-point IDEAL. *Magn Reson Med*. 2008; 59:521–527. [PubMed: 18306404]
12. Longo R, Pollesello P, Ricci C, et al. Proton MR spectroscopy in quantitative in vivo determination of fat content in human liver steatosis. *J Magn Reson Imaging*. 1995; 5:281–285. [PubMed: 7633104]
13. Longo R, Ricci C, Masutti F, et al. Fatty infiltration of the liver. Quantification by 1H localized magnetic resonance spectroscopy and comparison with computed tomography. *Invest Radiol*. 1993; 28:297–302. [PubMed: 8478169]
14. McPherson S, Jonsson JR, Cowin GJ, et al. Magnetic resonance imaging and spectroscopy accurately estimate the severity of steatosis provided the stage of fibrosis is considered. *J Hepatol*. 2009; 51:389–397. [PubMed: 19505740]
15. Thomsen C, Becker U, Winkler K, Christoffersen P, Jensen M, Henriksen O. Quantification of liver fat using magnetic resonance spectroscopy. *Magn Reson Imaging*. 1994; 12:487–495. [PubMed: 8007779]
16. Borra RJ, Salo S, Dean K, et al. Nonalcoholic fatty liver disease: rapid evaluation of liver fat content with in-phase and out-of-phase MR imaging. *Radiology*. 2009; 250:130–136. [PubMed: 19017926]
17. Bydder M, Yokoo T, Hamilton G, et al. Relaxation effects in the quantification of fat using gradient echo imaging. *Magn Reson Imaging*. 2008; 26:347–359. [PubMed: 18093781]
18. Fishbein MH, Gardner KG, Potter CJ, Schmalbrock P, Smith MA. Introduction of fast MR imaging in the assessment of hepatic steatosis. *Magn Reson Imaging*. 1997; 15:287–293. [PubMed: 9201675]
19. Guiu B, Petit JM, Loffroy R, et al. Quantification of liver fat content: comparison of triple-echo chemical shift gradient-echo imaging and in vivo proton MR spectroscopy. *Radiology*. 2009; 250:95–102. [PubMed: 19092092]
20. He T, Gatehouse PD, Kirk P, Mohiaddin RH, Pennell DJ, Firmin DN. Myocardial T-2* measurement in iron-overloaded thalassemia: an ex vivo study to investigate optimal methods of quantification. *Magn Reson Med*. 2008; 60:350–356. [PubMed: 18666131]
21. Hussain HK, Chenevert TL, Londy FJ, et al. Hepatic fat fraction: MR imaging for quantitative measurement and display—early experience. *Radiology*. 2005; 237:1048–1055. [PubMed: 16237138]
22. Kawamitsu H, Kaji Y, Ohara T, Sugimura K. Feasibility of quantitative intrahepatic lipid imaging applied to the magnetic resonance dual gradient echo sequence. *Magn Reson Med Sci*. 2003; 2:47–50. [PubMed: 16210819]
23. Kovanlikaya A, Guclu C, Desai C, Becerra R, Gilsanz V. Fat quantification using three-point dixon technique: in vitro validation. *Acad Radiol*. 2005; 12:636–639. [PubMed: 15866138]
24. Levenson H, Greensite F, Hoefs J, et al. Fatty infiltration of the liver: quantification with phase-contrast MR imaging at 1.5 T vs biopsy. *AJR Am J Roentgenol*. 1991; 156:307–312. [PubMed: 1898804]
25. Liu CY, McKenzie CA, Yu H, Brittain JH, Reeder SB. Fat quantification with IDEAL gradient echo imaging: correction of bias from T(1) and noise. *Magn Reson Med*. 2007; 58:354–364. [PubMed: 17654578]

26. Mitchell DG, Kim I, Chang TS, et al. Fatty liver. Chemical shift phase-difference and suppression magnetic resonance imaging techniques in animals, phantoms, and humans. *Invest Radiol.* 1991; 26:1041–1052. [PubMed: 1765436]
27. O'Regan DP, Callaghan MF, Wylezinska-Arridge M, et al. Liver fat content and T2*: simultaneous measurement by using breathhold multiecho MR imaging at 3.0 T—feasibility. *Radiology.* 2008; 247:550–557. [PubMed: 18349314]
28. Pacifico L, Celestre M, Anania C, Paolantonio P, Chiesa C, Laghi A. MRI and ultrasound for hepatic fat quantification: relationships to clinical and metabolic characteristics of pediatric nonalcoholic fatty liver disease. *Acta Paediatr.* 2007; 96:542–547. [PubMed: 17306008]
29. Qayyum A, Goh JS, Kakar S, Yeh BM, Merriman RB, Coakley FV. Accuracy of liver fat quantification at MR imaging: comparison of out-of-phase gradient-echo and fat-saturated fast spin-echo techniques—initial experience. *Radiology.* 2005; 237:507–511. [PubMed: 16244259]
30. Reeder SB, McKenzie CA, Pineda AR, et al. Water-fat separation with IDEAL gradient-echo imaging. *J Magn Reson Imaging.* 2007; 25:644–652. [PubMed: 17326087]
31. Reeder SB, Robson PM, Yu H, et al. Quantification of hepatic steatosis with MRI: the effects of accurate fat spectral modeling. *J Magn Reson Imaging.* 2009; 29:1332–1339. [PubMed: 19472390]
32. Rinella ME, McCarthy R, Thakrar K, et al. Dual-echo, chemical shift gradient-echo magnetic resonance imaging to quantify hepatic steatosis: implications for living liver donation. *Liver Transpl.* 2003; 9:851–856. [PubMed: 12884199]
33. Schuchmann S, Weigel C, Albrecht L, et al. Non-invasive quantification of hepatic fat fraction by fast 1.0, 1.5 and 3.0 T MR imaging. *Eur J Radiol.* 2007; 62:416–422. [PubMed: 17267159]
34. Yoshimitsu K, Kuroda Y, Nakamuta M, et al. Noninvasive estimation of hepatic steatosis using plain CT vs. chemical-shift MR imaging: significance for living donors. *J Magn Reson Imaging.* 2008; 28:678–684. [PubMed: 18777551]
35. Yu H, McKenzie CA, Shimakawa A, et al. Multiecho reconstruction for simultaneous water-fat decomposition and T2* estimation. *J Magn Reson Imaging.* 2007; 26:1153–1161. [PubMed: 17896369]
36. Yu H, Shimakawa A, McKenzie CA, Brodsky E, Brittain JH, Reeder SB. Multiecho water-fat separation and simultaneous R2* estimation with multifrequency fat spectrum modeling. *Magn Reson Med.* 2008; 60:1122–1134. [PubMed: 18956464]
37. Hernando D, Hines CD, Yu H, Reeder SB. Addressing phase errors in fat-water imaging using a mixed magnitude/complex fitting method. *Magn Reson Med.* 2012; 67:638–644. [PubMed: 21713978]
38. Yu H, Shimakawa A, Hines CD, et al. Combination of complex-based and magnitude-based multiecho water-fat separation for accurate quantification of fat-fraction. *Magn Reson Med.* 2011; 66:199–206. [PubMed: 21695724]
39. Yu H, Shimakawa A, McKenzie CA, et al. Phase and amplitude correction for multi-echo water-fat separation with bipolar acquisitions. *J Magn Reson Imaging.* 2010; 31:1264–1271. [PubMed: 20432366]
40. Pineda AR, Reeder SB, Wen Z, Pelc NJ. Cramer-Rao bounds for three-point decomposition of water and fat. *Magn Reson Med.* 2005; 54:625–635. [PubMed: 16092102]
41. Reeder, S.; Hines, C.; Yu, H.; McKenzie, C.; Brittain, J. Relationship between proton-density fat-fraction, true fat concentration for in vivo fat quantification with magnetic resonance imaging. *Proceedings of the 19th Annual Meeting of ISMRM; Montreal, Canada.* 2011. (abstract 805)
42. Reeder, S.; Hines, C.; Yu, H.; McKenzie, C.; Brittain, J. On the definition of fat-fraction for in vivo fat quantification with magnetic resonance imaging. *Proceedings of the 17th Annual Meeting of ISMRM; Honolulu, Hawaii.* 2009. (abstract 207)
43. Hu, H.; Li, Y.; Nagy, T.; Goran, M.; Nayak, K. Quantification of absolute fat mass: a validation study between chemical-shift MRI and chemical analysis. *Proceedings of the 20th Annual Meeting of ISMRM; Melbourne, Australia.* 2012. (abstract 1266)
44. Hines CD, Agni R, Roen C, et al. Validation of MRI biomarkers of hepatic steatosis in the presence of iron overload in the ob/ob mouse. *J Magn Reson Imaging.* 2012; 35:844–851. [PubMed: 22127834]

45. Hines CD, Yu H, Shimakawa A, McKenzie CA, Brittain JH, Reeder SB. T1 independent, T2* corrected MRI with accurate spectral modeling for quantification of fat: validation in a fat-water-SPIO phantom. *J Magn Reson Imaging*. 2009; 30:1215–1222. [PubMed: 19856457]
46. Hines CD, Frydrychowicz A, Hamilton G, et al. T(1) independent, T(2) (*) corrected chemical shift based fat-water separation with multi-peak fat spectral modeling is an accurate and precise measure of hepatic steatosis. *J Magn Reson Imaging*. 2011; 33:873–881. [PubMed: 21448952]

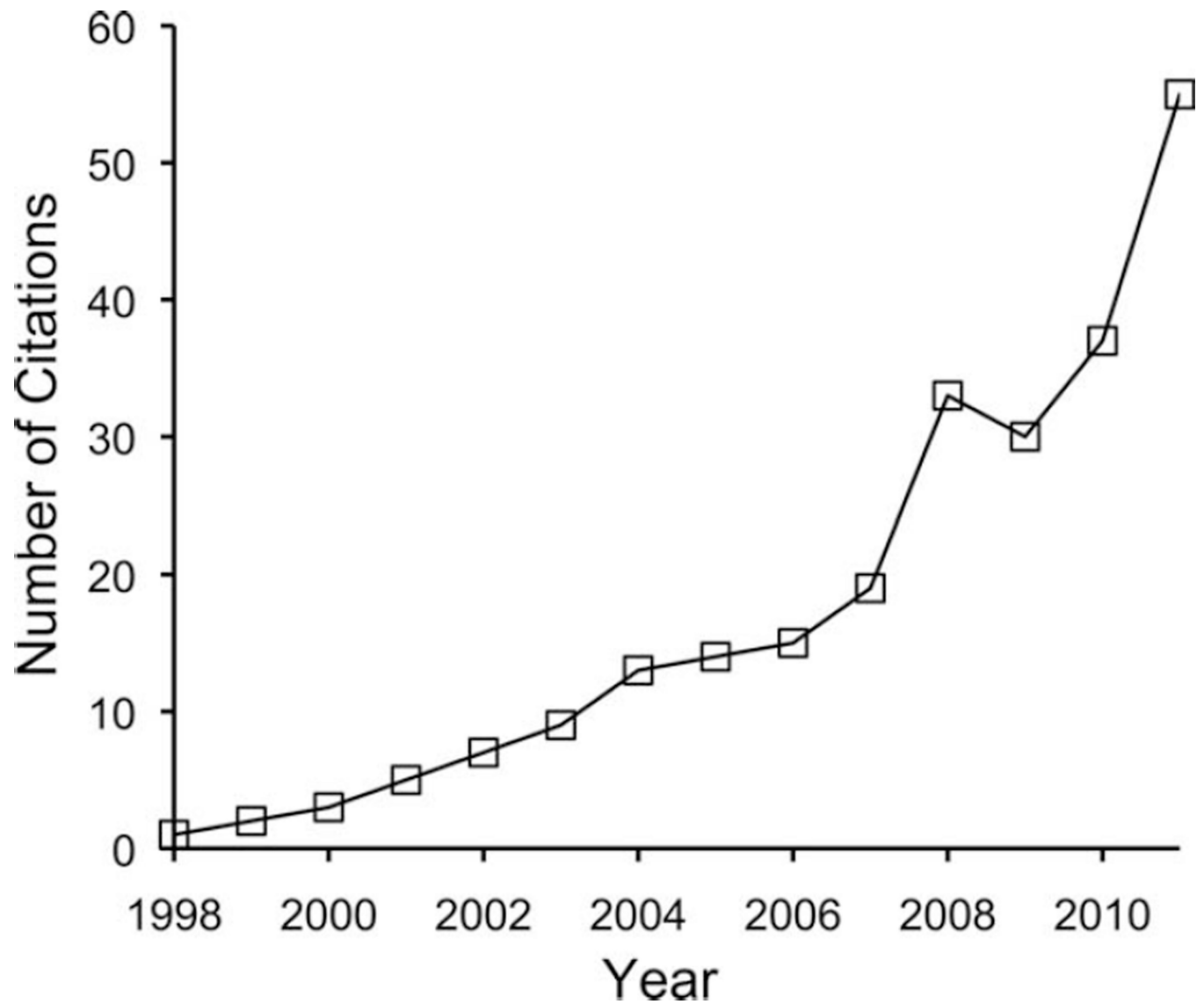


Figure 1.

The number of citations on fat quantification with magnetic resonance has grown steadily in the past decade. This pubmed.org search, limited to 1998 through 2011, was conducted on April 20, 2012, using key words “fat quantification magnetic resonance”.